

# ORGNZ-02 LAB: Understanding Grail Buckets - Hands-on Exercises

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**Series:** ORGNZ — Organize Data: Buckets, Segments, Security | **Notebook:** 2 of 10 | **Type:** LAB | **Created:** February 2026 | **Last Updated:** 04/26/2026

## Overview

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This lab notebook contains 5 hands-on exercises extracted from **ORGNZ-02: Understanding Grail Buckets**. Complete the lecture notebook first, then work through these exercises to reinforce the concepts with real DQL queries against your Dynatrace environment.

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## Table of Contents

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- [Exercise 1: Bucket Administration Permissions](#)
  - [Exercise 2: Bucket Administration Permissions](#)
  - [Exercise 3: Bucket Administration Permissions](#)
  - [Exercise 4: Querying Data from Buckets](#)
  - [Exercise 5: Querying Data from Buckets](#)
  - [Lab Summary](#)
- 

## Prerequisites

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| Requirement | Details                                                  |
|-------------|----------------------------------------------------------|
| Completed   | ORGNZ-02: Understanding Grail Buckets (lecture notebook) |

| Requirement           | Details                                               |
|-----------------------|-------------------------------------------------------|
| Dynatrace Environment | SaaS tenant with Grail enabled                        |
| Permissions           | logs.read , metrics.read , entities.read , spans.read |

## Exercise 1: Bucket Administration Permissions

# ORGNZ-02: Understanding Grail Buckets

**Series:** ORGNZ — Organize Data: Buckets, Segments, Security | **Notebook:** 2 of 10 | **Created:** January 2026 | **Last Updated:** 04/26/2026

**Buckets** are logical storage containers in Dynatrace Grail where records are stored. Think of a bucket as a folder in a filesystem. Use buckets to group telemetry data that naturally belongs together, such as data from the same region, environment, or with the same sensitivity classification.

| Requirement | Details |  
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- 1. What Are Buckets?
- 2. Supported Data Types (Tables)
- 3. Bucket Limits
- 4. Query Constraints
- 5. Default Buckets
- 6. Querying Bucket Information
- 7. Querying Data from Buckets
- 8. Bucket Characteristics
- 9. Bucket Naming Rules
- 10. When to Create Custom Buckets

| Permission                                       | Purpose                    |
|--------------------------------------------------|----------------------------|
| <code>storage:bucket-definitions:read</code>     | View bucket configurations |
| <code>storage:bucket-definitions:write</code>    | Create and modify buckets  |
| <code>storage:bucket-definitions:delete</code>   | Remove buckets             |
| <code>storage:bucket-definitions:truncate</code> | Truncate bucket data       |

By the end of this notebook, you will:

- Understand bucket fundamentals and supported data types
- Know bucket limits and query constraints
- Query bucket information using DQL
- Recognize when custom buckets are needed

Buckets organize data to meet performance, compliance, and retention requirements:

| Function                        | Description                                                 |
|---------------------------------|-------------------------------------------------------------|
| <b>Data Grouping</b>            | Organize large datasets into meaningful segments            |
| <b>Streamlined Organization</b> | Records that should be handled together are stored together |
| <b>Query Performance</b>        | Reduced queried dataset improves query speed                |
| <b>Retention Management</b>     | Each bucket has its own retention policy                    |
| <b>Access Control</b>           | Bucket-level IAM policies for team isolation                |
| <b>Cost Attribution</b>         | Enable data cost tracking by organizational unit            |

Each bucket is associated with exactly **one data type**:

| Data Type        | Default Bucket                 | Description                  |
|------------------|--------------------------------|------------------------------|
| <b>logs</b>      | <code>default_logs</code>      | Log records from all sources |
| <b>metrics</b>   | <code>default_metrics</code>   | Metric data points           |
| <b>spans</b>     | <code>default_spans</code>     | Distributed trace spans      |
| <b>events</b>    | <code>default_events</code>    | Platform events              |
| <b>bizevents</b> | <code>default_bizevents</code> | Business events              |

| Data Type              | Default Bucket          | Description             |
|------------------------|-------------------------|-------------------------|
| <b>security_events</b> | default_security_events | Security-related events |

**Important:** You cannot mix data types in the same bucket. A logs bucket only stores logs.

| Limit                           | Value                    | Notes                               |
|---------------------------------|--------------------------|-------------------------------------|
| Maximum buckets per environment | 80                       | Default limit; can request increase |
| Typical capacity                | Up to 5 TB/day per table | With default bucket limit           |

| Metric                    | Value      | Impact                          |
|---------------------------|------------|---------------------------------|
| Optimal ingest per bucket | ~1 TB/day  | Best query performance          |
| Acceptable ingest         | 1-3 TB/day | Limited query window            |
| Maximum ingest            | 3 TB/day   | Performance degrades above this |

| Limit             | Value                           |
|-------------------|---------------------------------|
| Minimum retention | 1 day                           |
| Maximum retention | 3,657 days (~10 years + 1 week) |

Understanding query limits is critical for bucket planning:

| Limit                    | Value  | Impact                               |
|--------------------------|--------|--------------------------------------|
| Maximum data scanned     | 500 GB | Limits queryable time window         |
| Maximum records returned | 1,000  | Use aggregations for larger datasets |
| Maximum response payload | 1 MB   | Large result sets may be truncated   |

The 500 GB scan limit determines how far back you can query:

| Daily Ingest | Queryable Window |
|--------------|------------------|
| 500 GB/day   | ~24 hours        |

| Daily Ingest | Queryable Window |
|--------------|------------------|
| 1 TB/day     | ~12 hours        |
| 3 TB/day     | ~4 hours         |
| 10 TB/day    | ~1 hour          |

**Implication:** High-volume buckets may only allow querying recent data. Use aggregations, time filters, and strategic bucket partitioning to work within these limits.

Dynatrace provides default buckets with varying retention:

| Default Bucket                       | Data Type       | Default Retention | Notes                       |
|--------------------------------------|-----------------|-------------------|-----------------------------|
| <code>default_logs</code>            | logs            | 35 days           | Standard log retention      |
| <code>default_metrics</code>         | metrics         | 15 months         | Extended for trend analysis |
| <code>default_spans</code>           | spans           | 10 days           | Short-term APM data         |
| <code>default_events</code>          | events          | 35 days           | Platform events             |
| <code>default_bizevents</code>       | bizevents       | 35 days           | Business events             |
| <code>default_security_events</code> | security_events | 35 days           | Security events             |

**Note:** Query your environment's `dt.system.buckets` to verify current retention settings.

```
// List all buckets with retention and status
fetch dt.system.buckets
| fields name, display_name, dt.system.table, retention_days,
estimated_uncompressed_bytes
| sort dt.system.table asc, name asc
```

## Exercise 2: Bucket Administration Permissions

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```
// Query logs from default bucket (implicit)
fetch logs, from:-1h
| limit 10
```

## Exercise 3: Bucket Administration Permissions

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```
// Query logs from a specific custom bucket
fetch logs, from:-1h, bucket:{"default_logs"}
| limit 10
```

## Exercise 4: Querying Data from Buckets

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```
// Query from multiple buckets simultaneously
fetch logs, from:-1h, bucket:{"default_logs", "audit_logs", "security_logs"}
| limit 100
```

## Exercise 5: Querying Data from Buckets

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```
// Filter by bucket within query
fetch logs, from:-1h
| filter dt.system.bucket == "default_logs"
| filter loglevel == "ERROR"
| limit 50
```

## Lab Summary

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You have completed 5 hands-on exercises for **ORGNZ-02: Understanding Grail Buckets**.

## Exercises Completed

- [ ] Exercise 1: Bucket Administration Permissions
- [ ] Exercise 2: Bucket Administration Permissions
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- [ ] Exercise 5: Querying Data from Buckets

## Next Steps

Continue with **ORGNZ-03** for the next notebook.

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*This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.*